Barnes–Wall Lattice Automorphism Groups
The index-two correction and direct lattice-preservation verification

Computational verification note

17 July 2026

Abstract

Away from the exceptional rank $m = 3$, the Barnes–Wall automorphism group is an index-two subgroup of the real Clifford group. We prove the resulting twisted sigma–MGF identity and verify it exhaustively in the concrete models $BW_2 = \mathbb{Z}^2$ and $BW_4 = D_4$. The constructed groups have orders 8 and 1152 and agree exactly with the Clifford matrices preserving the specified lattice bases. All 16 coefficient comparisons pass.

1 The index-two identity

Assume $m \neq 3$ and write

$$B_m = \mathrm{Aut}(BW_{2^m}) = \ker \varepsilon_m \quad \text{inside} \quad \mathcal{C}_m,$$

where $\varepsilon_m : \mathcal{C}_m \to \{\pm 1\}$ is the nontrivial quotient character. Define the twisted series

$$\mathfrak{M}^{\varepsilon_m}_{\mathcal{C}_m,m}(\mathbf{t}) = \frac{1}{|\mathcal{C}_m|} \sum_{g \in \mathcal{C}_m} \varepsilon_m(g) \prod_j \det(I - t_j g)^{-1}.$$

Since $\mathbf{1}_{B_m}(g) = (1 + \varepsilon_m(g))/2$ and $|B_m| = |\mathcal{C}_m|/2$, direct substitution proves

$$\boxed{\mathfrak{M}_{B_m,m} = \mathfrak{M}_{\mathcal{C}_m,m} + \mathfrak{M}^{\varepsilon_m}_{\mathcal{C}_m,m}.} \tag{1}$$

For $W_\tau = \bigotimes_i \mathrm{Sym}^{\tau_i} V_m$, coefficient extraction equivalently gives

$$\dim W_\tau^{B_m} = \dim W_\tau^{\mathcal{C}_m} + \dim \mathrm{Hom}_{\mathcal{C}_m}(\varepsilon_m, W_\tau).$$

The second nonnegative summand is precisely the lattice-sensitive sector.

The structural equality $\mathrm{Aut}(BW_{2^m}) = \ker \varepsilon_m \leq \mathcal{C}_m$ and the exceptional behavior at $m = 3$ are external inputs from the cited Barnes–Wall/Clifford-group literature. Equation (1) is proved conditionally from this index-two statement, and the verifier checks the concrete ranks $m = 1, 2$.

2 Concrete lattice models

For $m = 1$ we use $BW_2 = \mathbb{Z}^2$. Its automorphism group is the signed permutation group of order 8, equal to E_1, while the real Clifford group has order 16.

For $m = 2$ we use

$$BW_4 = D_4 = \{x \in \mathbb{Z}^4 : \textstyle\sum_i x_i \equiv 0 \pmod 2\}.$$

A basis is given by the columns of

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Starting with E_2, tensor Hadamard $H \otimes H$, CNOT, and controlled-Z, matrix closure gives a subgroup of order 1152. Independently, for each of the 2304 matrices $g \in \mathcal{C}_2$ the verifier tests whether $L^{-1}gL \in GL(4, \mathbb{Z})$. Exactly the same 1152 matrices pass. Thus the subgroup identification uses lattice preservation, not only an expected order. Every matrix in both closed groups is also checked to have real entries and satisfy $g^{\mathsf{T}} g = I$.

3 Direct versus twisted averages

For each partition τ, the direct side averages $\prod_i h_{\tau_i}(\operatorname{spec} g)$ over B_m. On the right side, the same integrand is averaged first with weight 1 and then with weight $\varepsilon_m(g)$ over all of $\mathcal{C}_m$. Membership in the independently identified lattice stabilizer determines the sign.

m	$\lvert B_m\rvert$	τ	direct	Clifford	twisted	sum
1	8	(4)	2	1	1	2
1	8	(8)	3	2	1	3
1	8	$(4, 2)$	4	2	2	4
2	1152	(4)	1	1	0	1
2	1152	(6)	2	1	1	2
2	1152	(8)	3	2	1	3
2	1152	$(4, 2)$	3	2	1	3

These rows exhibit a real distinction: the first tested one-row twisted contribution occurs in degree 4 for B_1 but only degree 6 for B_2. The full sweep tests eight partitions at each rank, including mixed powers; all 16 instances of (1) pass.

4 Scope and reproduction

No $m = 3$ conclusion is drawn: that rank is explicitly exceptional to the index-two statement. Ranks $m \geq 4$ are covered algebraically by (1), but exhaustive closure is not a reasonable verification strategy because the groups grow rapidly.

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.venv/bin/python for_this_guy/barnes_wall_lattice_automorphism_groups/
  barnes_wall/verification.py
.venv/bin/marimo edit for_this_guy/barnes_wall_lattice_automorphism_groups/
  barnes_wall/notebook.py
```

Join each wrapped command onto one line. Background on the group chain and the exceptional rank appears in G. Nebe, E. Rains, and N. J. A. Sloane, *The invariants of the Clifford groups*, https://arxiv.org/abs/math/0001038.